Reversed polarity: a control law for holding a maintained epigenetic state—and why it is not what bulk aging measures

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A cell that maintains its epigenetic identity against copy noise can, in principle, do so in two regimes. In the *disordered* regime it pays continuously to correct each locus, and information still decays toward a floor (the aging of the companion paper [1]). In an *ordered* regime—where read–write feedback couples neighboring loci strongly enough—the youthful configuration becomes a broken-symmetry free-energy well, held at near-zero maintenance cost, and drifting away must cross a collective barrier. We formalize this "reverse-polarity" regime with the same governing law as the companion paper, $\tau_0 \dot{M} = JM - T\,\mathrm{artanh}(M) + h_{\mathrm{ext}}$, give the economics of when collective protection beats per-locus pumping, and test it interventionally in simulation. Imposing strong local consensus (neighbor agreement, no stored reference) on a two-dimensional lattice—the rule is a symmetric neighbourhood majority vote, a standard 2D low-temperature memory—sustains an ordered phase. At $p = 0.10$, the aligned domain retains a strict global majority in every 2D replicate through the 4,000-division horizon, while matched one-dimensional coupled lineages lose that majority. The intervention is bounded: it requires $\geq$ 2-dimensional connectivity (1D cannot order) and it fails above a critical copy noise $p_c \approx 0.22$. A first-passage sweep strengthens the dimensional claim: at matched size and $p = 0.10$, all 1D replicates fail (median 43–64 divisions across sizes), while none of 32 2D replicates fails by 4,000 divisions; near criticality ($p = 0.20$), median 2D lifetime grows from 318 to 2,725 divisions as L grows from 8 to 32. (An earlier draft reported a counter-intuitive "size negative"—larger domains melting more near criticality—as a third bound; a replicated size sweep does not reproduce it, so it is withdrawn.) We map three biological levers (lower noise T, raise coupling J, re-nucleate order) onto Polycomb/heterochromatin read–write feedback, note the oncogenesis/rigidity frontier that bounds over-imposed order, and specify a synthetic-domain size-sweep experiment that would validate the physics. The contribution is a control-theoretic characterization of a maintained epigenetic state—its equilibrium, its barrier, its failure modes—not a claim that natural aging is this melting reversed.

I. FRAMING

This paper gives a control law for the maintained-state dynamics of the same epigenetic order parameter M, and under the same governing law, as the companion paper—but in the opposite *regime*. The companion paper concludes human blood sits in the *disordered* regime ($T > J$, $\mathcal{N} = J/T_{\mathrm{eff}} < 1$): the law reduces to relaxation-to-floor, there is no collective barrier, and the measured aging is cell-composition mixing plus per-locus decay with no within-cell melting signature at composition-inert sites. Here we explore the *ordered* regime ($T < J$, $\mathcal{N} > 1$), where a broken-symmetry barrier exists and can be engineered. This is the honest decoupling: not "different variables" (a physicist rightly reads that as evasion) but *one equation, two regimes separated by* $\mathcal{N} = 1$. We do *not* claim the aging the companion paper measures is a melting run backward—in the $\mathcal{N} < 1$ regime there is no ordered phase to have melted. We ask what a control law that *engineers* the $\mathcal{N} > 1$ regime could do; it is not a wet-lab reversal-of-aging claim. Every "arrests" below is bound to the simulation and its critical-noise regime.

II. ORDERED-PHASE PHYSICS: A MAINTAINED STATE AS A BROKEN SYMMETRY

The governing law's ordered limit ($J > T$, dimension ≥ 2) develops two free-energy wells at $\pm M_0$ separated by a barrier Δf. A domain sitting in the $+M_0$ well is a broken-symmetry memory: no stored reference tells it which well is "correct," but once committed it resists noise, because escaping requires the whole coupled domain to cross the barrier together. The Arrhenius escape time scales as $\tau \propto \exp(n\,\Delta f/T)$ in the coherent-flip picture, where n is the domain size—the naive reading in which bigger domains are exponentially safer. Section IV shows why that reading fails near criticality.

Two hard constraints frame the regime:

- *Dimensionality.* There is no ordered phase in 1D at finite temperature ($T_c = 0$): a 1D chain of coupled loci cannot hold a broken-symmetry state, because a single domain wall costs finite energy and entropy proliferates it. Order requires $\geq$ 2-dimensional connectivity—a testable prediction confirmed in Sec. IV (1D consensus fails, 2D holds). The stronger statement is that no *simple local* 1D rule stores a bit against noise: a reliable 1D memory is not strictly impossible (Gács's positive-rates automaton [7] is non-ergodic in 1D) but requires a baroque hierarchical self-simulating construction no biological read–write loop possesses, whereas a 2D neighbourhood majority does it with a simple local rule.

- *The one-bit bound.* What an ordered domain protects is its coarse *order parameter*—the $\sim$1 bit of "which well"—not an arbitrary per-locus codeword. Consensus dynamics smooth away per-locus detail; they preserve the collective alignment. This is the von Neumann one-bit reliability bound [2] made physical: a maintained state stores one robust bit per domain, cheaply, and cannot cheaply store fine-grained identity.

III. ECONOMICS: WHEN COLLECTIVE PROTECTION BEATS PER-LOCUS PUMPING

A cell has two ways to resist decay. It can *pump* each locus individually (kinetic reinforcement rate r per site, the disordered-regime maintenance of the companion paper), or it can *couple* loci into a collective ordered domain that holds many sites with one shared apparatus. The trade-off is set by how the apparatus cost scales with the number of protected loci. Costing both strategies per locus held stable over D divisions (in Landauer resets [5], each maintenance write $\geq k_B T \ln 2$): direct per-locus correction costs

$$\mathrm{cost}_{\mathrm{pump}} = D\, p_{\mathrm{eff}}, \tag{1}$$

while an ordered domain in which one collective barrier stabilizes b correlated loci with n_c apparatus units per domain, each turning over with probability γ per division, costs

$$\mathrm{cost}_{\mathrm{ordered}} = D\left(\frac{n_c\,\gamma}{b} + e^{-B/T}\right), \tag{2}$$

where the residual $e^{-B/T}$ is the (negligible, deep-well) spontaneous-flip cost. Equating the two and dropping the residual, the ordered phase wins iff $n_c\gamma/b < p_{\mathrm{eff}}$, i.e. the critical apparatus turnover is

$$\gamma^* = \frac{p_{\mathrm{eff}}\, b}{n_c}. \tag{3}$$

The entire content is two independence assumptions. (i) $n_c \perp b$: the win is the $1/b$ amortization, which requires *sub-linear* apparatus—one collective read–write field, not a reader/writer bolted to every site. If the machinery is per-locus ($n_c \propto b$) the amortization cancels, γ^* pins at p_{eff}, and the ordered phase collapses into per-locus pumping (the "pump the pump" failure mode). (ii) Barrier depth B drops out of Eq. (3) algebraically, but is paid *inside* n_c: depth buys Arrhenius longevity, not economics. Both costs scale with the lifetime division count D ($\sim 10^2$ for blood stem cells [12]), which therefore cancels from the crossover. Grounding the rest (Fig. 1, right): the per-locus error is the measured DNMT1 fidelity, $p_{\mathrm{eff}} \approx 5 \times 10^{-2}$ (maintenance methylation is $\sim$95% accurate per CpG per division [10, 11]); b is a domain's loci, read off its physical extent at $\sim$200 bp per nucleosome—heterochromatin runs from $\sim$1 kb nanodomains to Mb blocks ($b \sim 10$–10^3) [13, 14], Polycomb tens to hundreds of kb ($b \sim 10^2$–10^3) [15]. The two remaining constants are *not* directly measured: n_c (apparatus units per domain) is a model abstraction whose *structure* the biology fixes—sub-linear for heterochromatin/Polycomb, which nucleate and spread from a few sites, versus $n_c \propto b$ for DNA-methylation maintenance, where UHRF1/DNMT1 read and rewrite each hemimethylated CpG [15]—but whose value it does not; and γ is the probability a domain fails to re-establish before the next S-phase (*not* the $\sim$2-fold replicative dilution, which is normally repaired), which the slow post-replication recovery of H3K9me3/H3K27me3 renders non-negligible [16]. Under these unmeasured constants the discriminating verdict among the three systems moves by more than an order of magnitude, and at the measured p_{eff} it collapses entirely—all three fall on the ordered-wins side. We therefore claim only the robust part: a large, durable, genuinely collective domain (heterochromatin) beats per-locus pumping, while *ranking* Polycomb against DNA-methylation maintenance awaits a measurement of n_c and γ. This is a cost claim only: a domain carries $\sim$1 identity bit, so the win is cheap redundancy across coupled loci, not tighter identity.

IV. THE INTERVENTION TEST: ENGINEERED CONSENSUS ARRESTS DRIFT, BUT IS BOUNDED

We test reverse polarity as an *intervention*: impose strong local consensus and ask whether the youthful state becomes the free equilibrium so that aging is the expensive, uphill move. The protected object is an aligned domain (the $\sim$1 bit), started all-aligned; each division applies binary-symmetric copy noise (flip probability p) then optional

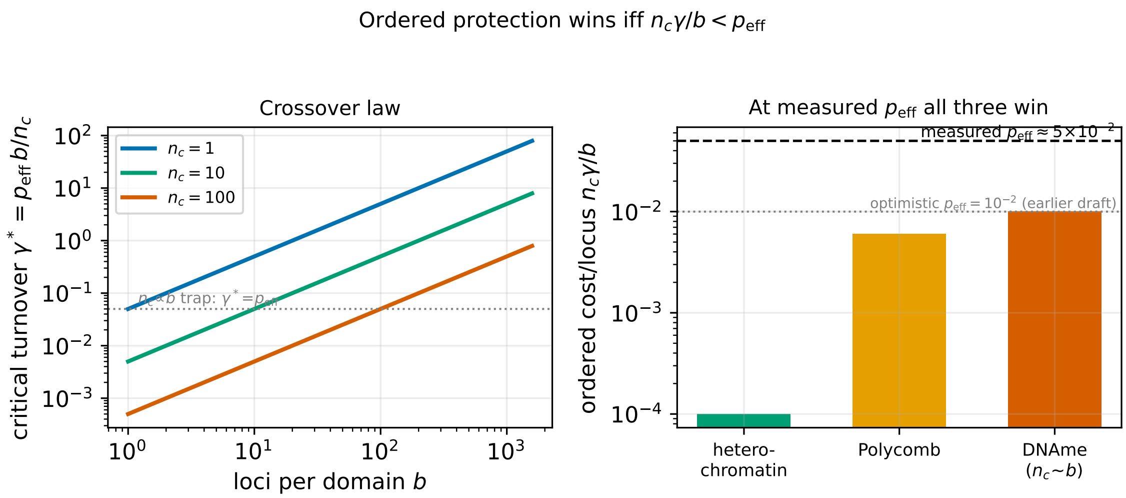

FIG. 1. The maintenance-economics crossover $\gamma^* = p_{\rm eff}\, b/n_c$ (Eq. (3)). *Left:* critical apparatus turnover vs loci per domain, for several apparatus sizes n_c; collective barriers (large b) and cheap apparatus (small n_c) raise γ^*, while the per-site diagonal $n_c \propto b$ pins it at $p_{\rm eff}$. *Right:* the three chromatin systems as illustrative points on the same surface. At the measured maintenance error $p_{\rm eff} \approx 5 \times 10^{-2}$ all three fall on the ordered-wins side; the earlier win/borderline/lose ranking was an artifact of an optimistic $p_{\rm eff} = 10^{-2}$, and only heterochromatin's large-domain margin is robust to the unmeasured n_c and γ.

reinforcement. Retention $R(t) = \langle \mathbf{1}[\text{state}_t = \text{state}_0] \rangle$; $R = 1$ fully young, $R = 0.5$ fully melted. Reinforcement is neighbor consensus only—never comparison to a stored codeword—so the dynamics contain no backup; matching to the initial state is a *measurement*, not a correcting force (the load-bearing invariant). This 2D torus rule is a *symmetric neighbourhood majority vote* (each site adopts the majority of its 3×3 window). Symmetric majority voting on a 2D lattice is a standard low-temperature memory: below a critical noise it sustains a symmetry-broken ordered phase and shrinks minority (aged) droplets by surface tension, which is the mechanism we simulate. (It is *not* Toom's north–east–centre rule [6]; that asymmetric eroder is provably robust even to *biased* noise, a stronger property the symmetric rule lacks in the infinite-size limit. We return to the bias question, and why it does not change the biological conclusion, below.)

TABLE I. Retention over divisions at sub-critical noise $p = 0.10$ (4096 loci, 150 divisions; seed 20260722).

arm	mechanism	$R[10]$	$R[50]$	$R[150]$
baseline	no reinforcement	0.545	0.496	0.501
1D consensus	ring neighbor majority	0.780	0.533	0.507 (fails, $T_c = 0$)
2D consensus	torus neighbor majority (reverse polarity)	1.000	0.999	1.000 (arrested)

The 2D-coupled domain holds $R \approx 1$ over this 150-division run (Table I) while baseline collapses to the melted floor within ~10 divisions (front-loaded, as the core picture predicts) and 1D—which briefly slows decay—still relaxes to ≈ 0.5. This is "aging made expensive," built from local consensus alone, no backup.

a. Dose-response. Steady R (2D, mean of the last 40 of 150 divisions) stays near 1 for small p ($R = 0.99$ at $p = 0.15$, 0.96 at $p = 0.18$) and collapses through a maintained-to-melted transition at $p_c \approx 0.22$ ($R = 0.57$ at $p = 0.22$, 0.49 at $p = 0.28$). The engineered barrier is finite; above p_c, noise wins regardless of coupling. Our p_c is a measured value for this specific symmetric majority rule, not a claim to match any particular published constant.

b. Size dependence near criticality (a withdrawn negative). An earlier draft reported, from a *single* lineage per size, that retention *declines* with domain size near p_c ($R = 0.973 \to 0.935 \to 0.844 \to 0.874$ for $L = 8, 16, 32, 64$ at $p = 0.20$) and read it as area-dominated droplet nucleation ("bigger is not safer"). We flagged it as needing a size sweep. That sweep, run at 40 replicates per cell across the near-critical band ($p = 0.16$–0.21; `sim_size_negative_stress.py`), does *not* reproduce it. The size-slope of retention is flat or slightly *positive* at every p (at $p = 0.20$, $+0.0010$ per unit L, bootstrap 95% CI $[+0.0002, +0.0019]$), the single-run "$R(L{=}64) < R(L{=}8)$" call is a coin flip across replicates (55–68%), and at long horizons the *smallest* domain is the least stable while larger domains hold ($L = 8, 16, 32, 64 \to R = 0.51, 0.54, 0.72, 0.79$ at $p = 0.20$, $D = 2000$)—the opposite of area-nucleation, and consistent with finite-size rounding of the transition. The original declining curve was a single-lineage fluctuation near a noisy critical point. We therefore withdraw the size-negative. This does not establish a "bigger is safer" law either: at the accessible sizes an ordered $L = 64$ domain does not fully melt within the simulated horizon, so the sweep bounds the claim rather than realizing droplet nucleation. The dimensionality requirement and the p_c cliff are unaffected.

c. First-passage lifetime scaling. We therefore replaced fixed-horizon retention as the size diagnostic with first loss of the stored domain bit (`scripts/sim_memory_lifetime_scaling.py`; 32 independent replicates per cell, matched $n = L^2$ loci, right-censored at 4,000 divisions). At $p = 0.10$, all 1D rings fail; median first-passage time is 43–64 divisions across $L = 8$–32, whereas no 2D domain fails by the horizon. At $p = 0.20$, median 2D lifetime rises from 318 ($L = 8$) to 614 ($L = 16$) to 2,725 divisions ($L = 32$), and the failure fraction falls from 100% to 94% to 63%. At the estimated transition ($p = 0.22$), every 2D replicate fails and the size benefit is small (median 84–176 divisions). These 32-replicate, right-censored values are point estimates: finite-horizon evidence for a size-stabilized 2D memory below the transition, not proof of infinite lifetime.

d. Verdict. Reverse polarity works as an intervention *in the model*: below p_c, high-connectivity consensus sustains an ordered phase holding the youthful domain at $R \approx 1$, using only local consensus and no backup. But it is bounded and conditional—needs $\geq$ 2-dimensional connectivity, fails above p_c, costs durable collective apparatus, and over-imposing order is the rigidity/oncogenesis failure mode (Sec. VI).

e. Biased copy error (withdrawn claim). Epigenetic copy error is directional—hemimethylation and 5mC loss on replication are not symmetric spin flips—so a maintenance rule must hold a bit against *biased* noise. An earlier draft argued, from the (mistaken) Toom identification, that this forces maintenance to be a non-equilibrium, detailed-balance-violating eroder, because a symmetric majority rule was expected to fail under bias. We tested that directly (`scripts/sim_bias_robustness.py`): an all-ones domain copied through a strongly loss-biased channel and reinforced by the symmetric majority vote *holds* the identity bit far past where a genuine 3-cell Toom NEC rule collapses (symmetric retention ≈ 1 at a loss rate $p_{10} = 0.08$ where Toom is already at 0). Toom's provable robustness to biased noise is an infinite-size/time property that does not manifest at simulatable scales, where the larger symmetric neighbourhood simply has the higher noise threshold. Our simulations therefore do *not* support the claim that maintenance must be non-equilibrium to resist biased copy error; the symmetric maintained phase already tolerates it in the accessible regime, so we withdraw that argument. The asymptotic Toom/positive-rates distinction may still matter in principle, but it is neither needed for nor demonstrated in the biological regime we simulate.

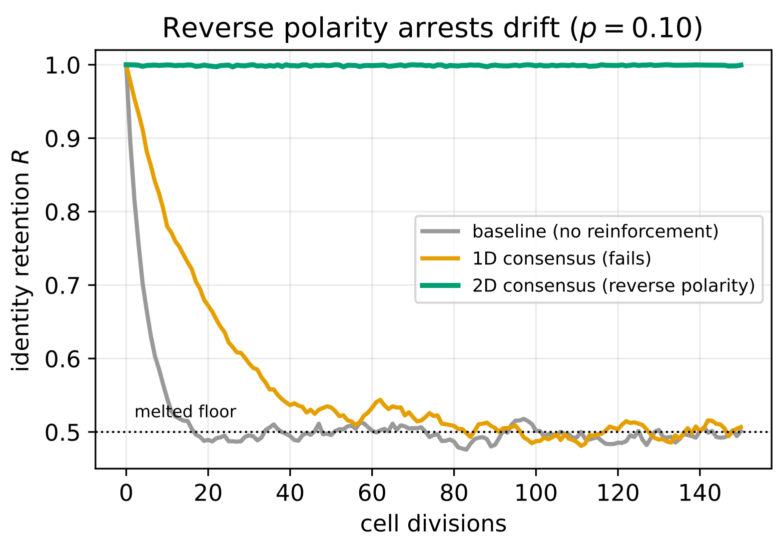

FIG. 2. Identity retention over divisions at $p = 0.10$ (seed 20260722). 2D neighbor consensus (reverse polarity) holds $R \approx 1$ over this 150-division run; baseline collapses to the melted floor within ~10 divisions; 1D consensus only briefly slows decay and still relaxes to ≈ 0.5.

V. INFORMATION VERSUS POWER: A HARD CEILING AND A KINETIC FLOOR

Ordered protection does not evade the information limits of the companion paper; it relocates them. There is a hard *information* ceiling [4]: no maintenance—collective or per-locus—can raise $I(\text{birth};\text{state})$ without an external

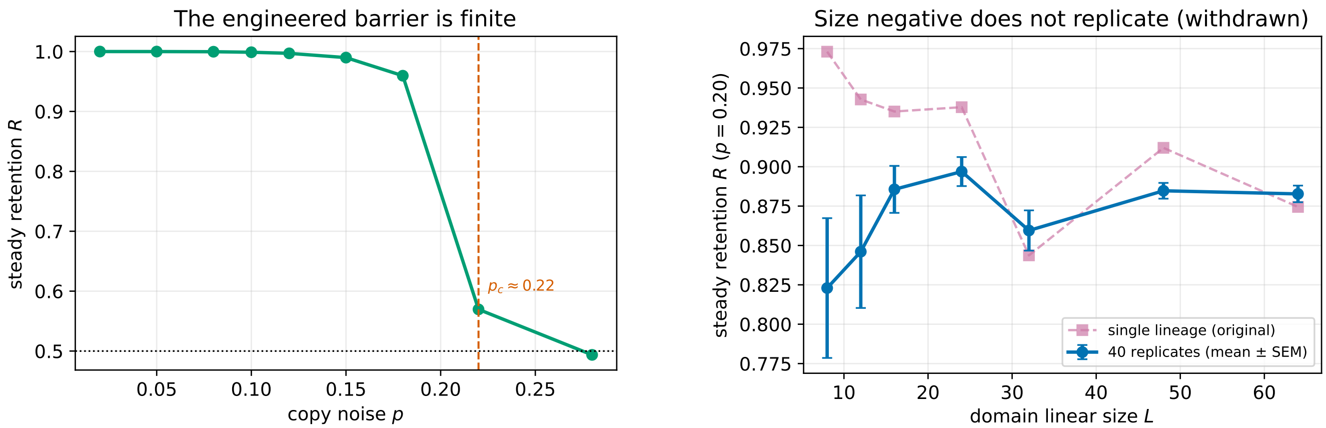

FIG. 3. Bounds on the intervention. *Left:* dose-response—steady retention holds near 1 up to a maintained-to-melted transition at $p_c \approx 0.22$. *Right:* the *withdrawn* size negative. A single lineage (dashed) suggested retention falls with domain size near criticality ($p = 0.20$); 40 replicates (mean $\pm$ SEM) do not reproduce a decline—if anything the smallest domain is least stable. The original curve was a single-lineage fluctuation; the area-nucleation "bigger melts more" claim is withdrawn.

reference (the $\mathbb{Z}_2$ symmetry). Reverse polarity buys *retention* of the $\sim$1 collective bit, not *restoration* of lost per-locus identity. There is also a *kinetic* floor distinct from Landauer's thermodynamic one: holding the barrier requires a minimum reinforcement rate to outrun copy noise (the Eigen threshold [3] $r > p_{\mathrm{raw}}(1-p^*)/p^*$), and this floor is set by rates, not by the $k_B T \ln 2$ erasure cost. The information wall (if $T(\rho) > J$, the ordered phase is unreachable at any metabolic power) is the hard boundary of what any intervention can achieve.

VI. BIOLOGICAL REALIZATION AND ITS FRONTIER

The model's J (collective read–write coupling) and T (effective copy noise) map onto real chromatin feedback. Polycomb (H3K27me3) and constitutive heterochromatin (H3K9me3) are read–write systems: a mark recruits the enzyme that writes the same mark on neighbors— exactly the neighbor-consensus coupling that produces J. In the companion paper's reader/paper framing these are the three places to act—fix the reader, protect the paper, or rewrite it from an outside exemplar—giving three distinct interventional levers, each with a different mechanism and risk:

1. *Lower T*—improve copy fidelity / reduce the effective per-division flip rate. Broad but weak; approaches the information wall asymptotically.

2. *Raise J*—strengthen read–write feedback at identity loci (the reverse-polarity lever). Locus-selective; the intervention of Sec. IV.

3. *Re-nucleate*—re-establish an ordered domain that has already melted, from a boundary or seed. This is the only lever that *restores* rather than *retains*, and—consistently with the companion paper—it requires a reference (a seed/boundary the domain did not itself contain), i.e. an h_{ext}-like input.

a. The frontier. Over-imposed order is pathological. A locus locked into a collective well cannot respond to legitimate developmental or environmental signals (rigidity), and inappropriately maintained domains are a route to the epigenetic dysregulation of oncogenesis. Ordered protection is therefore necessarily locus-selective: biology spends it on identity cores and cannot afford (or survive) genome-wide order. This bounds the therapeutic envelope: the levers must be targeted, reversible, and regulated, not global.

VII. DISCUSSION

a. What this is. A control-theoretic characterization of a maintained epigenetic state: its equilibrium (an ordered well), its stability (a finite collective barrier), its controllability (three levers), and its failure modes (dimensionality, critical noise, nucleation, rigidity). The reverse-polarity regime is real physics with clean, falsifiable structure—an instance of 2D majority-vote memory (fault-tolerant cellular-automaton theory) applied to epigenetic maintenance rather than a new mechanism. The contribution is the mapping (identity maintenance = a symmetry-broken reliable memory; aging = its sub-threshold failure), not the automaton itself.

b. What this is not (restating Sec. I). It is not a claim that natural blood aging is this melting run backward. The companion paper shows measured bulk aging is composition mixing plus per-locus relaxation, with no within-cell melting signature at composition-inert sites up to age 52 (in the elderly, a read-level discordance signal concentrated in Polycomb domains—the meltable phase mapped here—does appear, but bisulfite data cannot yet separate it from an aged-DNA artifact); the ordered phase here is the *same* order parameter M under the *same* law, in the $\mathcal{N} > 1$ regime that natural blood ($\mathcal{N} < 1$) does not occupy. The honest link is negative: *because* natural blood aging is in the disordered regime with no ordered phase to melt, a reverse-polarity intervention would be engineering an $\mathcal{N} > 1$ regime nature does not use here, not restoring one it lost.

c. The validating experiment. Synthesize an ordered chromatin domain (e.g. a targeted Polycomb read–write loop over a defined locus array), vary its size, and measure the age/division dependence of its retention. The model predicts (i) a 2D-connectivity requirement—a domain wired for higher-dimensional consensus holds, a 1D-wired one does not—and (ii) a critical-noise cliff: retention collapses as the effective copy noise crosses p_c. Both are sharp, opposite to a null of gradual size-independent decay, and testable by a connectivity/noise sweep. (An earlier draft added a third prediction, a near-critical size negative; a replicated *in-silico* sweep withdrew it, so we do not carry it to the bench.) Flagged as future wet-lab work; this paper stops at the theory and simulation that motivate it.

VIII. METHODS

a. Dynamics. Binary-symmetric copy channel (flip probability p) plus invariant-safe local consensus reinforcement (1D ring and 2D torus neighbor majority, radius 1, adoption strength 1.0), no stored reference.

b. Order parameter / retention. Aligned-domain start; $R(t) = \langle \mathbf{1}[\text{state}_t = \text{state}_0] \rangle$; steady R = mean of the last 40 of 150 divisions.

c. Free energy / barrier. $f(M) = -\frac{J}{2}M^2 - T\,s(M)$; barrier and order parameter from the mean-field module.

d. Code and reproducibility. All simulation code and automated tests are in the palimpsest repository. Run values use the seeds recorded by their scripts.

Appendix A: Assessment of claims

TABLE II. Claims and the strength of their support.

claim	evidence	strength
Ordered phase exists (2D), with size-stabilized lifetime	first-passage sim + mean-field	moderate
No order for a simple local 1D rule	sim + 1D no-T_c argument	strong
Engineered consensus arrests drift below p_c	intervention sim (symmetric majority)	strong
Maintenance robust to biased copy error at accessible scales	bias sweep (sym. vs Toom)	moderate
Near-critical size negative (nucleation $\propto$ area)	40-replicate sweep does not reproduce it	withdrawn
Maps to Polycomb/heterochromatin read–write feedback	mechanism analogy	speculative
Natural aging is reversible melting	—	not claimed

Appendix B: Open experiments

1. Regenerate all provisional figures and numbers from a frozen run.

2. Measure the two unmeasured economics constants—n_c (apparatus units per domain) and γ (per-division re-establishment failure)—directly. The other constants (p_{eff}, b, D) are now grounded in Fig. 1, and the crossover law itself (Eq. (3)) is exact; the missing measurement is what would let the three chromatin systems be ranked rather than only bounded.

3. Test whether the lifetime scaling survives biologically plausible geometry, asynchronous updates, and nonuniform coupling rather than the square synchronous torus used here.

4. Frame every claim as theory + simulation; do not present it as a therapy result.

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